

TRANSFORMATION BETWEEN CARTESIAN, CYLINDRICAL AND SPERICAL COORDINATION SYSTEM

Aim:

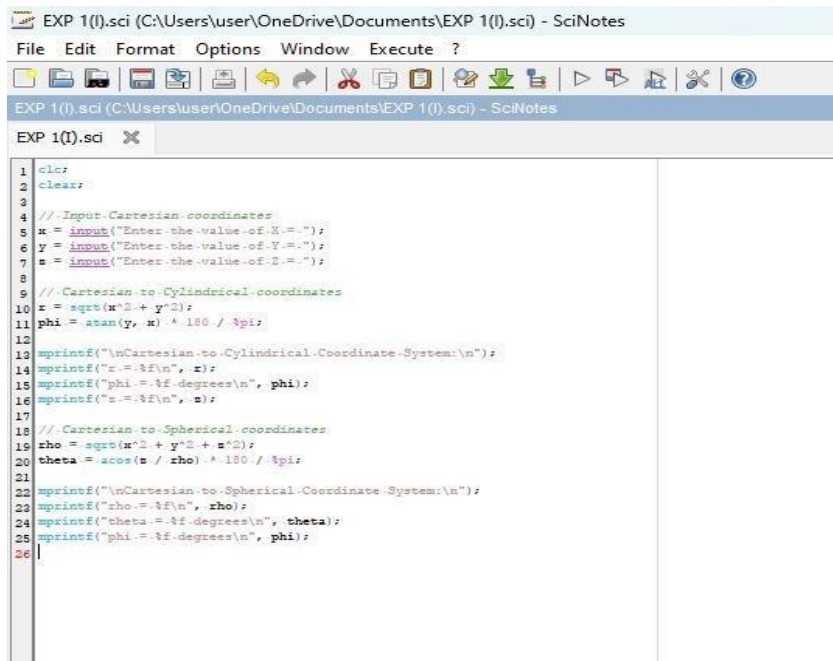
To write program for Transformation between Cartesian, Cylindrical and spherical coordinate system using SCILAB.

Software required:

- SCILAB version 6.0.1

Program:

Test case (i) Transformation of Cartesian coordinate points in to Cylindrical and spherical coordinate system Program:



```
EXP 1(i).sci (C:\Users\user\OneDrive\Documents\EXP 1(i).sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 1(i).sci (C:\Users\user\OneDrive\Documents\EXP 1(i).sci) - SciNotes
EXP 1(i).sci X
1 clr;
2 clear;
3
4 //Input Cartesian coordinates
5 x = input("Enter the value of X = ");
6 y = input("Enter the value of Y = ");
7 z = input("Enter the value of Z = ");
8
9 // Cartesian to Cylindrical coordinates
10 r = sqrt(x^2 + y^2);
11 phi = atan(y, x) * 180 / %pi;
12
13 fprintf("\nCartesian to Cylindrical Coordinate System:\n");
14 fprintf("r = %f\n", r);
15 fprintf("phi = %f-degrees\n", phi);
16 fprintf("z = %f\n", z);
17
18 // Cartesian to Spherical coordinates
19 rho = sqrt(x^2 + y^2 + z^2);
20 theta = acos(z / rho) * 180 / %pi;
21
22 fprintf("\nCartesian to Spherical Coordinate System:\n");
23 fprintf("rho = %f\n", rho);
24 fprintf("theta = %f-degrees\n", theta);
25 fprintf("phi = %f-degrees\n", phi);
26
```

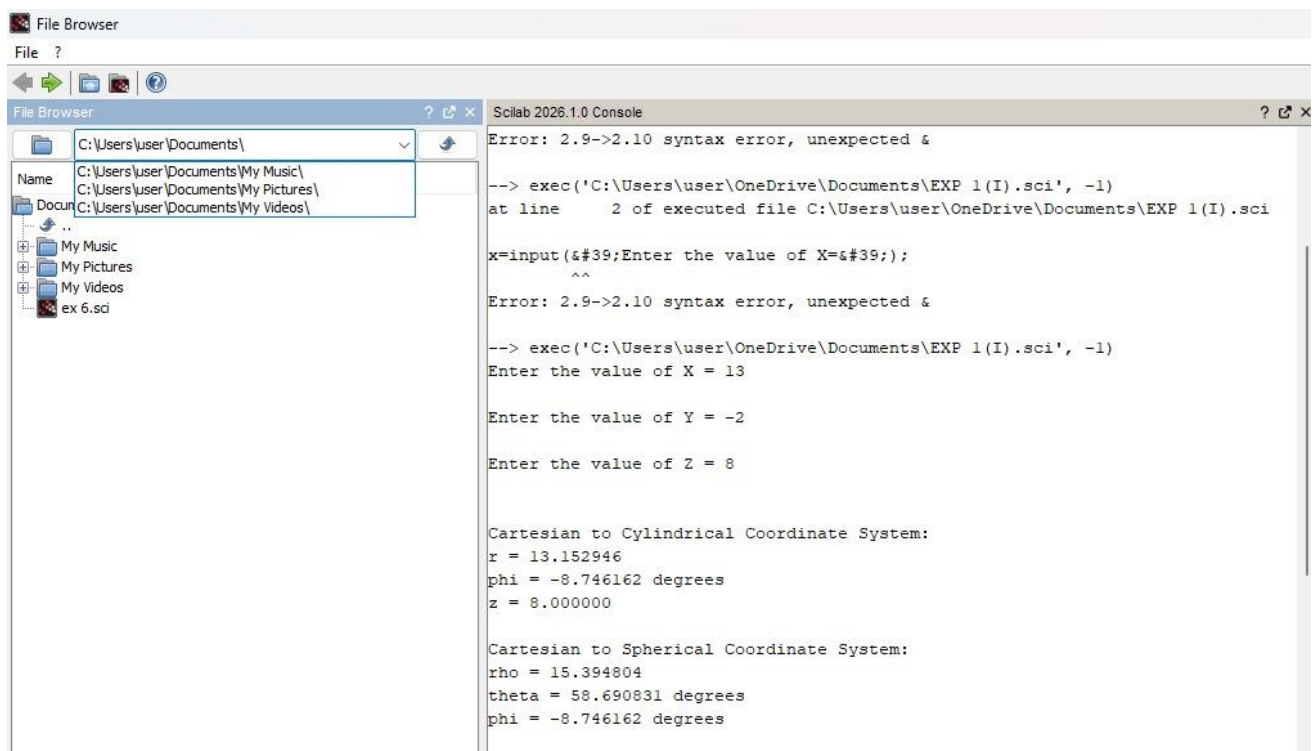
Output:-

Enter the value of $X=13$

Enter the value of $y=-2$

Enter the value of $z=8$ **Practical**

Output:



The screenshot displays the Scilab 2026.1.0 interface. On the left, the 'File Browser' window shows the directory structure of 'C:\Users\user\Documents\'. The main console window on the right shows the execution of a script. The script contains a syntax error on line 2, which is corrected in the subsequent execution. The script prompts the user to enter values for X, Y, and Z. The console output shows the results of the conversions from Cartesian to Cylindrical and Spherical coordinate systems.

```
File Browser
File ?
C:\Users\user\Documents\
Name
C:\Users\user\Documents\My Music\
C:\Users\user\Documents\My Pictures\
C:\Users\user\Documents\My Videos\
My Music
My Pictures
My Videos
ex 6.sci

Scilab 2026.1.0 Console
Error: 2.9->2.10 syntax error, unexpected &
--> exec('C:\Users\user\OneDrive\Documents\EXP 1(I).sci', -1)
at line      2 of executed file C:\Users\user\OneDrive\Documents\EXP 1(I).sci

x=input(&#39;Enter the value of X=&#39;);
    ^^
Error: 2.9->2.10 syntax error, unexpected &

--> exec('C:\Users\user\OneDrive\Documents\EXP 1(I).sci', -1)
Enter the value of X = 13

Enter the value of Y = -2

Enter the value of Z = 8

Cartesian to Cylindrical Coordinate System:
r = 13.152946
phi = -8.746162 degrees
z = 8.000000

Cartesian to Spherical Coordinate System:
rho = 15.394804
theta = 58.690831 degrees
phi = -8.746162 degrees
```

Theoretical output:

Theoretical :

Case (1) :

Rectangular $\rightarrow$ Cylindrical

$$\rho = \sqrt{x^2 + y^2} = \sqrt{(13)^2 + (-2)^2} = \sqrt{169 + 4}$$
$$= \sqrt{173} = 13.152$$

$$\boxed{\rho = 13.152}$$

$$\phi = \tan^{-1}(y/x)$$

$$= \tan^{-1}(-2/13) = \tan^{-1}(-0.153)$$

$$\boxed{\phi = -8.6987}$$

$$z = z$$

$$\boxed{z = 8}$$

Rectangular $\rightarrow$ Spherical

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{(13)^2 + (-2)^2 + (8)^2} = \sqrt{169 + 4 + 64}$$

$$r = \sqrt{237}$$

$$r = 15.39480432$$

$$\theta = \cos^{-1} \left(\frac{z}{\sqrt{x^2 + y^2 + z^2}} \right)$$

$$= \cos^{-1} \left(\frac{+8}{15.3948} \right)$$

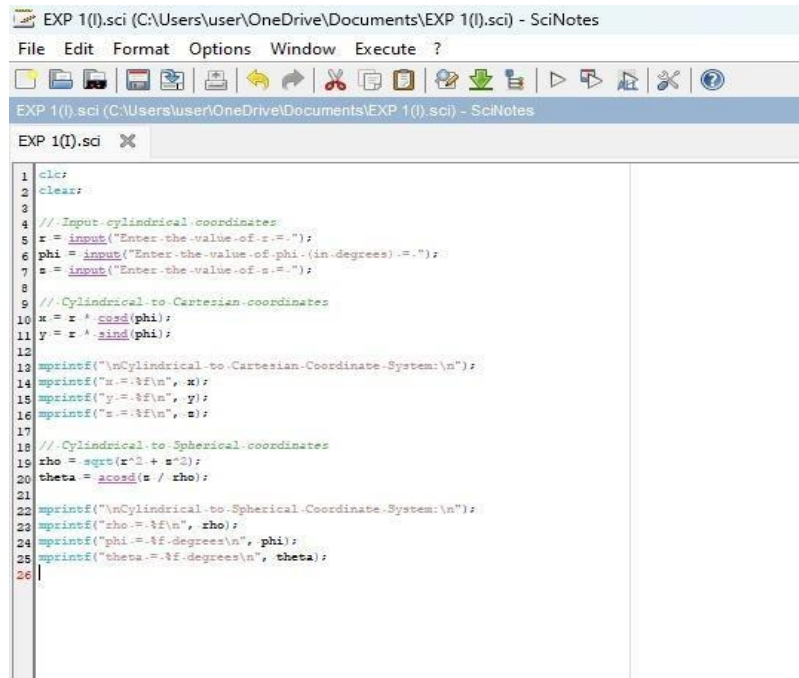
$$\theta = 58.69082$$

$$\phi = \tan^{-1}(y/x) = \tan^{-1}(-2/13)$$

$$\phi = -8.746162263$$

Test case (ii) Transformation of Cylindrical coordinate points into Cartesian and spherical coordinate system

Program:



```
1 clc;
2 clear;
3
4 // Input cylindrical coordinates
5 r = input("Enter the value of r:-");
6 phi = input("Enter the value of phi (in degrees):-");
7 z = input("Enter the value of z:-");
8
9 // Cylindrical to Cartesian coordinates
10 x = r * cos(phi);
11 y = r * sin(phi);
12
13 fprintf("\nCylindrical to Cartesian Coordinate System:\n");
14 fprintf("x = %f\n", x);
15 fprintf("y = %f\n", y);
16 fprintf("z = %f\n", z);
17
18 // Cylindrical to Spherical coordinates
19 rho = sqrt(r^2 + z^2);
20 theta = acos(z / rho);
21
22 fprintf("\nCylindrical to Spherical Coordinate System:\n");
23 fprintf("rho = %f\n", rho);
24 fprintf("phi = %f degrees\n", phi);
25 fprintf("theta = %f degrees\n", theta);
26
```

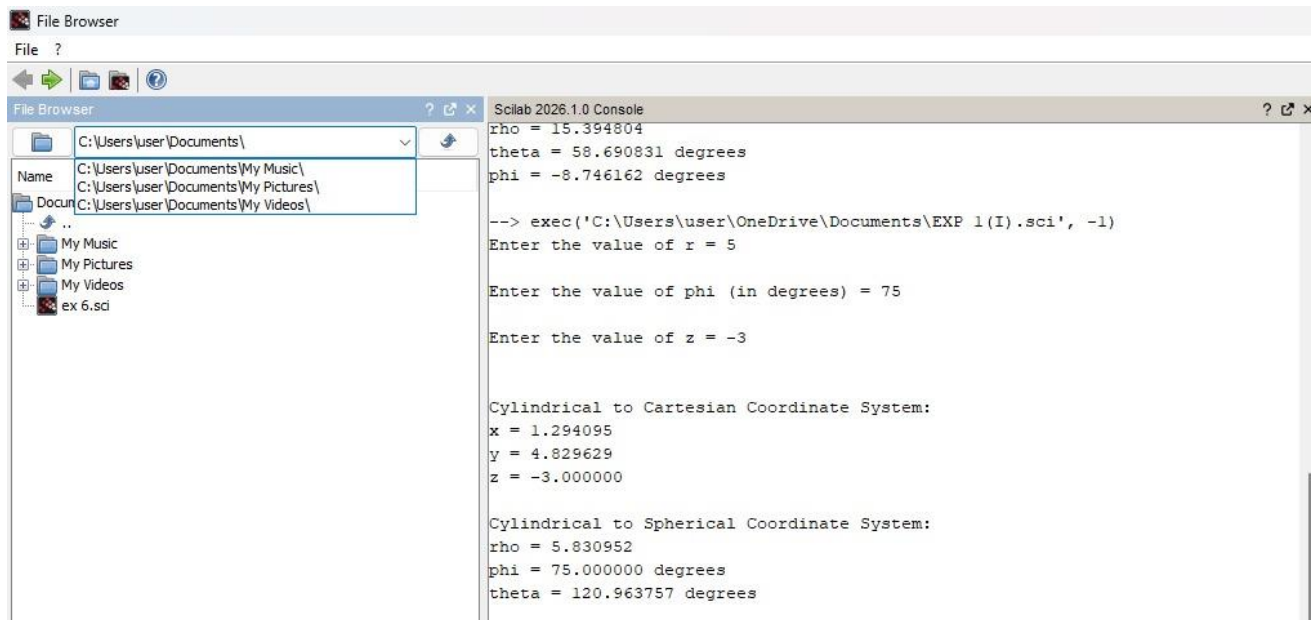
Output:-

Enter the value of r=5

Enter the value of pi=75

Enter the value of z= -3 **Practical**

Output:



The screenshot shows the Scilab 2026.1.0 interface. On the left is a File Browser window showing the directory structure of a user's documents. On the right is the Scilab 2026.1.0 Console window, which displays the output of a script. The script defines variables for rho, theta, and phi, and then performs two coordinate system conversions: Cylindrical to Cartesian and Cylindrical to Spherical.

```
File Browser
File ?
C:\Users\user\Documents\
Name
C:\Users\user\Documents\My Music\
C:\Users\user\Documents\My Pictures\
C:\Users\user\Documents\My Videos\
My Music
My Pictures
My Videos
ex 6.sci

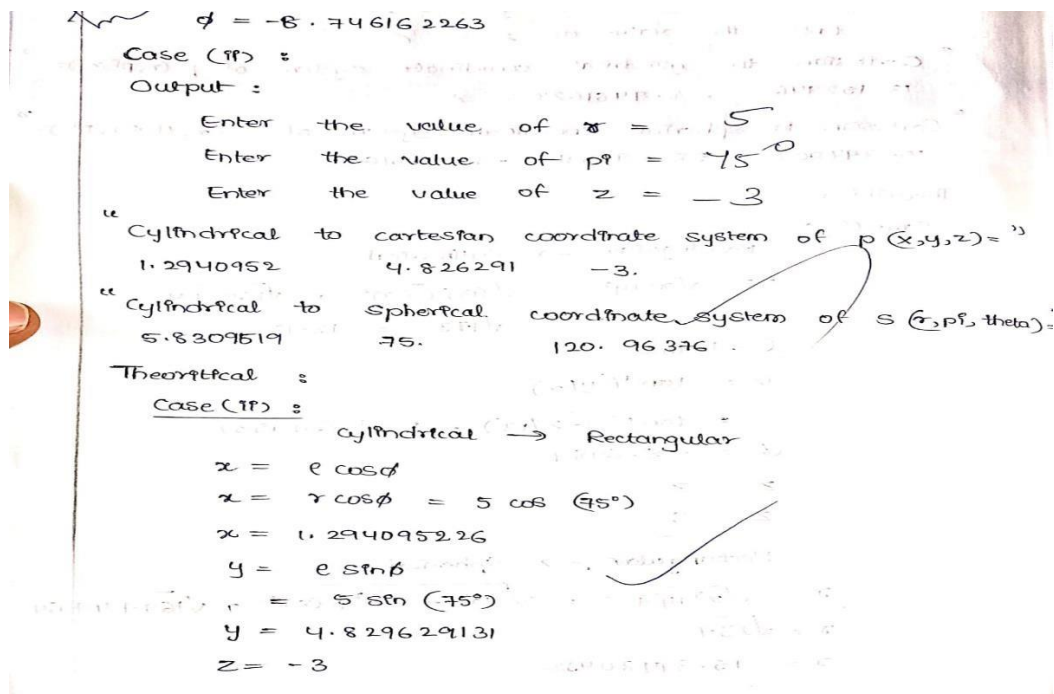
Scilab 2026.1.0 Console
rho = 15.394804
theta = 58.690831 degrees
phi = -8.746162 degrees

--> exec('C:\Users\user\OneDrive\Documents\EXP 1(I).sci', -1)
Enter the value of r = 5
Enter the value of phi (in degrees) = 75
Enter the value of z = -3

Cylindrical to Cartesian Coordinate System:
x = 1.294095
y = 4.829629
z = -3.000000

Cylindrical to Spherical Coordinate System:
rho = 5.830952
phi = 75.000000 degrees
theta = 120.963757 degrees
```

Theoretical output:



The handwritten text shows the theoretical output for the coordinate system conversion problem. It includes the input values for r, phi, and z, and the resulting Cartesian and Spherical coordinates. The Cartesian coordinates are x = 1.2940952, y = 4.8296291, and z = -3. The Spherical coordinates are rho = 5.8309519, phi = 75, and theta = 120.96376. The theoretical output is also shown, including the conversion formulas and the resulting values.

$\phi = -8.746162263$

Case (1):

Output:

Enter the value of $r = 5$

Enter the value of $\phi = 75^\circ$

Enter the value of $z = -3$

"Cylindrical to cartesian coordinate system of $p(x, y, z) =$ "

1.2940952 4.826291 -3.

"Cylindrical to Spherical coordinate system of $s(r, \phi, \theta) =$ "

5.8309519 75. 120.96376

Theoretical:

Case (1):

cylindrical $\rightarrow$ Rectangular

$x = r \cos \phi$

$x = 5 \cos 75^\circ$

$x = 1.294095226$

$y = r \sin \phi$

$y = 5 \sin 75^\circ$

$y = 4.829629131$

$z = -3$

Cylindrical to Spherical

$$r = \sqrt{\rho^2 + z^2} = \sqrt{(5)^2 + (-3)^2} = \sqrt{25+9} = \sqrt{34}$$

$$r = 5.830951895$$

$$\phi = \tan^{-1}(\rho/z) = \tan^{-1}(5/-3)$$

$$\theta = \cos^{-1}\left(\frac{z}{r}\right) = \cos^{-1}\left(\frac{-3}{5.830951895}\right)$$

$$\theta = 120.9693697$$

$$\phi = \phi$$

$$\phi = 75^\circ$$

Test case (iii) Transformation of spherical coordinate points into Cylindrical and Cartesian coordinate system Program:

```

EXP 1(i).sci (C:\Users\user\OneDrive\Documents\EXP 1(i).sci) - SciNotes
File Edit Format Options Window Execute ?
EXP 1(i).sci (C:\Users\user\OneDrive\Documents\EXP 1(i).sci) - SciNotes
EXP 1(i).sci X
1 clear;
2 clear;
3
4 // Input spherical coordinates
5 r = input("Enter the value of r = ");
6 phi = input("Enter the value of phi (in degrees) = ");
7 theta = input("Enter the value of theta (in degrees) = ");
8
9 // Spherical to Cartesian coordinates
10 x = r * sind(theta) * cosd(phi);
11 y = r * sind(theta) * sind(phi);
12 z = r * cosd(theta);
13
14 fprintf("\nSpherical to Cartesian Coordinate System:\n");
15 fprintf("x = %f\n", x);
16 fprintf("y = %f\n", y);
17 fprintf("z = %f\n", z);
18
19 // Spherical to Cylindrical coordinates
20 rho = r * sind(theta);
21 s_cyl = r * cosd(theta);
22
23 fprintf("\nSpherical to Cylindrical Coordinate System:\n");
24 fprintf("rho = %f\n", rho);
25 fprintf("phi = %f degrees\n", phi);
26 fprintf("s = %f\n", s_cyl);
27

```

Practical output:

```

Scilab 2026.0.1 Console
File Edit Control Applications ?
[Icons]
Scilab 2026.0.1 Console

Startup execution:
loading initial environment

--> exec('C:\Users\jasir\OneDrive\Documents\exp 1\iii_eme.sci', -1)
Enter the value of r=7

Enter the value of pi=120

Enter the value of theta=100

"spherical to cartesian coordinate system of p(x,y,z)="
5.6126821 4.0025328 -1.2155372
"spherical to cylindrical coordinate system of s(p,pi,z)="
-3.5445595 120. 5.6992668

-->

```

Theoretical output:

Case (iii) :

Output :

Enter the value of $r = 7$

Enter the value of $\phi = 120^\circ$

Enter the value of $\theta = 100^\circ$

"Spherical to cartesian coordinate system of $p(x,y,z) =$ "
5.6126821 4.0025328 -1.2155372

"Spherical to cylindrical coordinate system of $s(p,\pi,z) =$ "
-3.5445595 120. 5.6992668

Theoretical :

Case (iii) :

Spherical $\rightarrow$ Rectangular

$$x = r \cos \phi = 7 \cos(120^\circ) = r \sin \theta \cos \phi = 7 \sin(100^\circ) \cos(120^\circ)$$

$$x = 5.61268$$

$$y = r \sin \theta \sin \phi = 7 \sin(100^\circ) \sin(120^\circ)$$

$$y = 4.002$$

$$z = r \cos \theta = 7 \cos(100^\circ) = -1.21553724$$

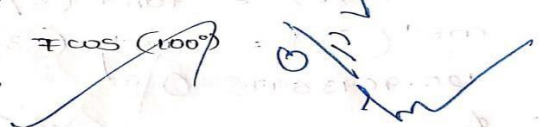
Spherical $\rightarrow$ Cylindrical

$$\rho = r \sin \theta = 7 \sin(100^\circ)$$

$$\rho = -3.54456$$

$$\phi = \phi = 120^\circ$$

$$z = r \cos \theta = 7 \cos(100^\circ)$$

$$z = 5.569926$$


Result:

Thus, the program was verified Transformation between Cartesian and Cylindrical coordinate system its

classified two categories (a) Cartesian to cylindrical and spherical coordinate system (b) Cylindrical to

cartesian and spherical coordinate system (c) spherical to cartesian and cylindrical coordinate system using SCILAB was successfully.

